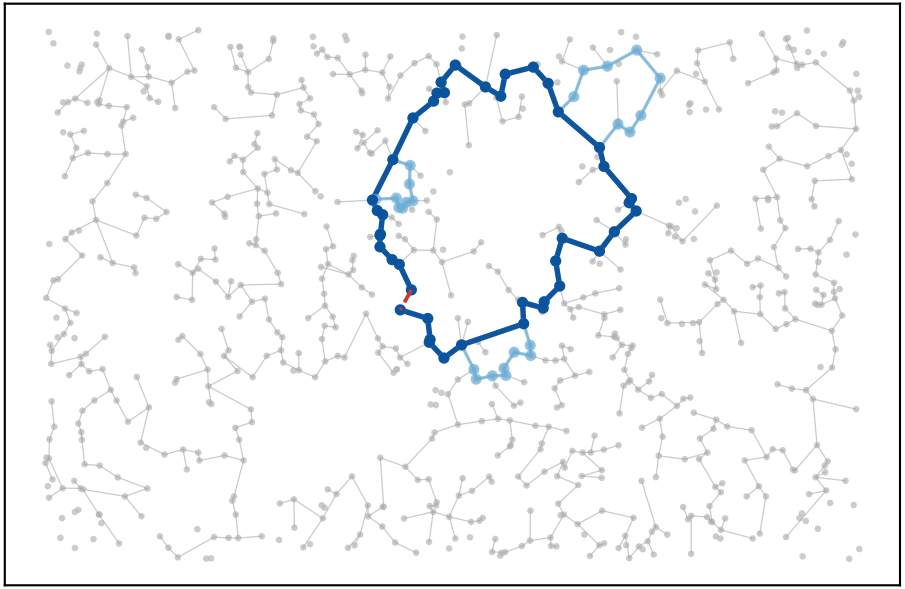
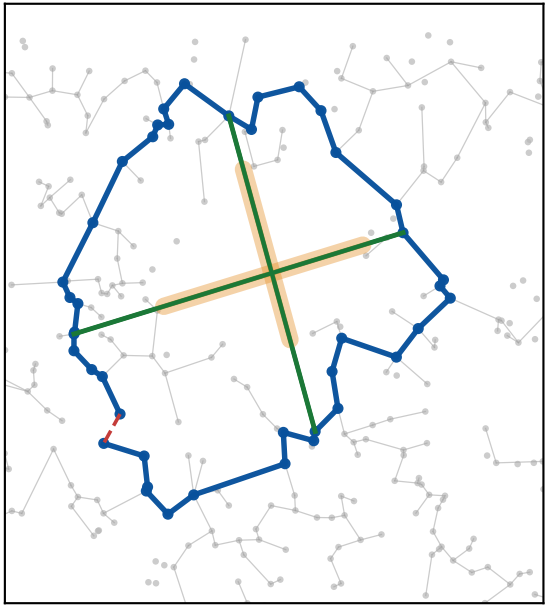
(a) Close-return path and shortcut cycle



(b) Obstruction chords and certified gaps